

Homework 8 Solutions

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September 16, 2022

Problem 17.1

- a) 1
- b) 1
- c) 9
- d) 9
- e) 0
- f) 1
- g) 1
- h) 0

Problem 17.2

$\{\{1, 2, 3\}\}, \{\{1, 2, 4\}\}, \{\{1, 2, 5\}\}, \{\{1, 2, 6\}\}, \{\{1, 3, 4\}\}, \{\{1, 3, 5\}\}, \{\{1, 3, 6\}\}, \{\{1, 4, 5\}\}, \{\{1, 4, 6\}\}, \{\{1, 5, 6\}\},$
 $\{\{2, 3, 4\}\}, \{\{2, 3, 5\}\}, \{\{2, 3, 6\}\}, \{\{2, 4, 5\}\}, \{\{2, 4, 6\}\}, \{\{2, 5, 6\}\}, \{\{3, 4, 5\}\}, \{\{3, 4, 6\}\}, \{\{3, 5, 6\}\}, \{\{4, 5, 6\}\}$

Problem 17.3

- a) 20
- b) -4320
- c) 0
- d) 20
- e) 0

Problem 17.4

There are $30 \cdot 29$ ways he can choose two socks to put on his feet with $\frac{30 \cdot 29}{2}$ ways of selecting just the two socks.

Problem 17.5

There are $19 + 18 + \dots + 1 = 190$ handshakes.

Problem 17.6

- a) $\binom{n}{k}$
- b) $\binom{n}{k} \cdot 2^{n-k}$

Problem 17.11

$$\binom{n}{4}$$

Problem 17.14

Proof.

$$2^n = \sum_{k=0}^n \binom{n}{k}$$

1. Consider the question "how many different subsets can be made from a set of size n ".
2. Observe that a set of all possible subsets is the power-set, and that the power-set has size 2^n .
3. Therefore 2^n is an answer to the question.
4. Observe that $\binom{n}{k}$ is how many size k subsets can be made of a set of size n .
5. Then the sum of all possible subsets from size 0 to n is $\sum_{k=0}^n \binom{n}{k}$.
6. Therefore $\sum_{k=0}^n \binom{n}{k}$ is an answer to the question.
7. Since 2^n and $\sum_{k=0}^n \binom{n}{k}$ both give a valid solution to the question, then $2^n = \sum_{k=0}^n \binom{n}{k}$.

□

Problem 17.19

Proof.

$$\binom{2n+2}{n+1} = \binom{2n}{n+1} + 2 \cdot \binom{2n}{n} + \binom{2n}{n-1}$$

1. Consider a $2n+2$ size element set with two particular elements a and b , and all the possible subsets of size $n+1$.
2. Observe that $\binom{2n+2}{n+1}$ is the count of all possible subsets of size $n+1$, and it remains to show that $\binom{2n}{n+1} + 2 \cdot \binom{2n}{n} + \binom{2n}{n-1}$ is also the count.
3. Observe that in counting the subsets, we can count all the ones that contain both a and b , then the ones that only contain a , then the ones that only contain b , then the ones that contain neither.
4. Observe that $\binom{2n}{n-1}$ counts all the subsets that contain both a and b since we remove two options and two choices.
5. Observe that $\binom{2n}{n}$ counts all the subsets that contain only a since we remove two options and only one choice.
6. Observe that $\binom{2n}{n}$ counts all the subsets that contain only b since we remove two options and only one choice.
7. Observe that $\binom{2n}{n+1}$ counts all the subsets that contain neither a nor b since we remove two options but no choices.
8. Then the total count is $\binom{2n}{n+1} + \binom{2n}{n} + \binom{2n}{n} + \binom{2n}{n-1} = \binom{2n}{n+1} + 2 \cdot \binom{2n}{n} + \binom{2n}{n-1}$.
9. Therefore $\binom{2n+2}{n+1} = \binom{2n}{n+1} + 2 \cdot \binom{2n}{n} + \binom{2n}{n-1}$.

□